ABASYN UNIVERSITY

Islamabad Campus

Mid-Term Examination – Fall-2026

Course Code: AI 216	Course Title: Programming for AI
Teacher's Name: Asad Hanif	Total Marks: 30
Date & Time: Nov 06, 2026 (09:00 am)	Total Time: 1 hour 30 minutes
Student's Name:	Reg. No:

1. Follow all the instructions written on answer booklet.
2. Students are required to read all questions carefully and ask for clarification (if required) in first 15 minutes of the start of the paper.

Note: Attempt all the questions. Mark of each question is written in front of the question.

QUESTION 1 [3+3 = 6 Marks] [CLO-1]

Version control and code tracing

Tasks

a) You edited **`labs/week04/task09_clean_code/main.py`**. Your repository already has a remote named origin and you are on the main branch. Write the **three Git commands** to stage only this file, commit it with the message "*Refactor prediction summary*", and push the commit to GitHub.

b) Trace the code below. **State the final value printed**, the values skipped by **`continue`**, and the value at which **`break`** stops the loop.

```
total = 0
for n in [3, 6, 9, 12]:
    if n % 2 != 0:
        continue
    if n > 10:
        break
    total += n
print(total)
```

QUESTION 2 [3+3 = 6 Marks] [CLO-1]

Response-time classification

A service has changed its response-time (in milliseconds) rules:

- Invalid if ms < 0;
- Fast if 0 ≤ ms ≤ 250;
- Acceptable if 250 < ms ≤ 600;
- Slow if ms > 600.

Tasks

a) Write a Python function named **`classify_response(ms)`** that returns the correct category string. Do not print inside the function.

b) For the values below, write a loop that builds a **dictionary named counts** with the number of Invalid, Fast, Acceptable, and Slow responses. Use your function; do not hard-code the counts.

```
latencies = [250, 251, 600, 601, -3]
```

QUESTION 3 [3+3 = 6 Marks] [CLO-1]

Find and fix two logic errors

```
def mean_nonnegative(values):
    total = 0
    for value in values:
        if value < 0:
            continue
        total = value
    return total / len(values)

print(mean_nonnegative([10, -5, 30]))
```

Tasks

a) Identify two different logic errors in the **`mean_nonnegative()`** function.

b) Rewrite the **`mean_nonnegative(values)`** function so that it averages only nonnegative values and returns None if there are no valid values. Do not change the input list.

QUESTION 4 [3+1+1+1 = 6 Marks] [CLO-1]

OOP, validation, exceptions, and modules

Tasks

a) Write a class `ConfidenceGate` whose constructor stores a threshold. Its accept(confidence) method must return True when confidence >= threshold and False otherwise. If confidence is outside the range 0 to 1 inclusive, the method must `raise ValueError`.

b) For the statement **`gate = ConfidenceGate(0.80),`** what are the return values of **`gate.accept(0.80)`** and **`gate.accept(0.79)`**.

c) Write a try/except block that calls **`gate.accept(1.20)`** and prints "Invalid confidence" when **`ValueError`** is raised.

d) If the class is saved in **`confidence_gate.py`**, write the ***import statement*** needed in **`main.py`**.

QUESTION 5 [2+2+2 = 6 Marks] [CLO-1]

Data structures for AI predictions

An AI classification model produces the predictions below. Each dictionary stores an ID, a predicted label, and a confidence score. Answer all parts using this data; do not change the original list.

```
predictions = [
    {"id": 201, "label": "spam", "confidence": 0.92},
    {"id": 202, "label": "ham", "confidence": 0.76},
    {"id": 203, "label": "spam", "confidence": 0.88},
    {"id": 204, "label": "review", "confidence": 0.81}
]
```

Tasks

a) Write a list comprehension to collect the IDs of predictions with confidence >= 0.85. Also show the resulting list.

b) Write a set comprehension to collect all unique predicted labels. Show the resulting set and explain in one sentence why a set is suitable.

c) Write a function **`count_labels(records)`** that uses a loop to return a dictionary showing how many times each label occurs. Also state the dictionary returned for the given predictions.

Best of Luck!